Session 4B: Errors, Files, CSV, and JSON

Trainer Reference Guide

Purpose and outcomes

Handle predictable failures with specific exceptions and persist structured user records with safe pathlib paths, CSV, and JSON.

Examples and exercises

- 01_exceptions.ipynb demonstrates try/except/else/finally and guides safe division.
- 02_files.py writes and reads a text report relative to its module, avoiding fragile current-directory assumptions.
- 03_csv_json.ipynb converts CSV rows to JSON-shaped dictionaries and distinguishes serialized text from Python objects.
- users.csv and sample_users.json provide realistic input data.
- exercise_01_safe_number.py is guided ValueError handling. exercise_02_json_users.py independently writes/loads JSON and handles missing or corrupt files.
- challenge_user_file_manager.py is Project 4: functions for load, save, add, list, search, and delete with persistence.

Trainer key points specific to this module

Catch only errors the code can explain or recover from. Keep try blocks narrow. Corrupt a copy of JSON live and inspect the traceback before handling it. Discuss encoding and why secrets do not belong in data files.

Portal connection and evidence

JSON becomes API payloads; SQLite later replaces the file while preserving CRUD concepts. Verify persistence across two program runs.

Delivery pattern

1. Explain the idea briefly and ask learners to predict a result.
2. Run the simple example, then connect it to an employee/user workflow.
3. Pair learners for the guided task; require an individual attempt next.
4. Use the challenge as the cumulative project increment.
5. Review the quiz and ask learners to explain, not only show, their code.

Trainer key points

- Keep live coding visible and deliberate. Normalize reading tracebacks.
- Ask what type and value each expression produces before running it.
- Let TODO files remain student-owned; open solutions only during review.
- Check boundary and failure cases, not only the happy path.
- End by tracing how the topic appears in the final portal.